

$$\int_0^{\frac{\pi}{4}} 32 \frac{2 - \sec^2 x}{\sec^2 x + 4 \sin^2 x} \log_2(\sin x + \cos x) dx$$

$$32 \int_0^{\frac{\pi}{4}} \frac{2 \sec^2 x - 2 \tan^2 x - \sec^2 x}{\sec^2 x + 4 \sin^2 x} \frac{\ln(\sin x + \cos x)}{\ln 2} dx$$

$$\frac{32}{\ln 2} \int_0^{\frac{\pi}{4}} \frac{\sec^2 x - 2 \tan^2 x}{\sec^2 x + 4 \sin^2 x} \ln(\sin x + \cos x) dx$$

$$\frac{32}{\ln 2} \int_0^{\frac{\pi}{4}} \frac{1 - 2 \sin^2 x}{4 \sin^2 x \cos^2 x + 1} \ln(\sin x + \cos x) dx$$

$$\frac{32}{\ln 2} \int_0^{\frac{\pi}{4}} \frac{\cos 2x}{1 + \sin^2 2x} \ln(\sin x + \cos x) dx$$

$$(\sin x + \cos x)^2 = \sin^2 x + 2 \sin x \cos x + \cos^2 x = 1 + \sin 2x$$

$$\sin x + \cos x = \pm \sqrt{1 + \sin 2x}$$

$$\sin x + \cos x \geq 0 \text{ st. } x \in [0, \frac{\pi}{4}]$$

$$\therefore \sin x + \cos x := \sqrt{1 + \sin 2x}$$

$$\frac{16}{\ln 2} \int_0^{\frac{\pi}{4}} \frac{\cos 2x}{1 + \sin^2 2x} \ln(1 + \sin 2x) dx$$

$$u := \sin 2x$$

$$du = 2 \cos 2x$$

$$\frac{8}{\ln 2} \int_0^1 \frac{\ln(1+u)}{1+u^2} du$$

$$u := \tan t$$

$$du = \sec^2 t dt$$

$$\frac{8}{\ln 2} \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan t)}{1+\tan^2 t} \sec^2 t dt$$

$$\frac{8}{\ln 2} \int_0^{\frac{\pi}{4}} \ln(1 + \tan t) dt$$

$$\frac{8}{\ln 2} \int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos t + \sin t}{\cos t}\right) dt$$

$$\frac{8}{\ln 2} \left(\int_0^{\frac{\pi}{4}} \ln(\cos t + \sin t) dt - \int_0^{\frac{\pi}{4}} \ln(\cos t) dt \right)$$

$$\sin(\alpha + \beta) = \cos \alpha \sin \beta + \sin \alpha \cos \beta$$

$$\beta := \frac{\pi}{4}$$

$$\sin\left(\alpha + \frac{\pi}{4}\right) = \cos \alpha \frac{\sqrt{2}}{2} + \sin \alpha \frac{\sqrt{2}}{2}$$

$$\cos \alpha + \sin \alpha = \sqrt{2} \sin\left(\alpha + \frac{\pi}{4}\right)$$

$$\frac{8}{\ln 2} \left(\int_0^{\frac{\pi}{4}} \ln\left(\sqrt{2} \sin\left(t + \frac{\pi}{4}\right)\right) dt - \int_0^{\frac{\pi}{4}} \ln(\cos t) dt \right)$$

$$\frac{8}{\ln 2} \left(\int_0^{\frac{\pi}{4}} \ln(\sin(t + \frac{\pi}{4})) dt + \frac{1}{2} \int_0^{\frac{\pi}{4}} \ln 2 dt - \int_0^{\frac{\pi}{4}} \ln(\cos t) dt \right)$$

$$w := t + \frac{\pi}{4}$$

$$dw = dt$$

$$\frac{8}{\ln 2} \left(\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \ln(\sin w) dw + \ln \frac{2}{2} t \Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \ln(\cos t) dt \right)$$

$$A := \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \ln(\sin w) dw$$

$$B := \int_0^{\frac{\pi}{4}} \ln(\cos t) dt$$

$$y := \frac{\pi}{2} - t$$

$$dy = -dt$$

$$B = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \ln(\cos(\frac{\pi}{2} - y)) dy$$

$$B=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\ln(\sin y)dy$$

$$\therefore A=B\wedge A-B=0$$

$$\frac{8}{\ln 2}\Big(A-B+\frac{\ln 2}{2}\frac{\pi}{4}\Big)$$

$$\frac{8}{\ln 2}\frac{\ln 2}{8}\pi$$

$$\pi$$